

MULTIMODALITY IMAGING OF SYNCHRONOUS BILATERAL AND IPSILATERAL BREAST CANCERS WITH DISTINCT HISTOLOGICAL SUBTYPES: A CASE SERIES

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BACKGROUND

Synchronous breast cancer is an uncommon presentation that may occur as bilateral or ipsilateral primary malignancies diagnosed within six months. **Distinct histological subtypes within synchronous tumours** present diagnostic and therapeutic challenges. **Multimodality imaging** is essential for lesion detection, radiologic-pathologic correlation, staging and treatment planning.

CASE PRESENTATION

Two cases of synchronous breast cancer with different histology are presented. A 59-year-old asymptomatic woman had bilateral screen-detected invasive carcinoma NST and mucinous carcinoma on mammography, ultrasound and CEM. A 46-year-old woman presented with a palpable lump and two ipsilateral lesions, confirmed as invasive carcinoma NST and mucinous carcinoma. Both had node-negative disease with clear margins and no distant metastases.

DISCUSSION

These cases highlight the **heterogeneous imaging appearances of synchronous breast cancers** and emphasize that lesions with **benign imaging features, particularly mucinous carcinoma, may coexist with more aggressive malignancies**. Independent assessment and biopsy of each suspicious lesion are essential. Mammography, ultrasonography and CEM provide complementary information for accurate diagnosis, staging and surgical planning.

IMAGING FEATURES

CASE 1: BILATERAL Synchronous tumour

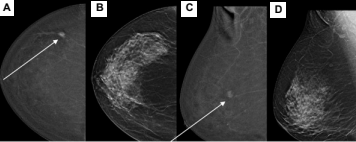


Figure 1A-D: Contrast-Enhanced Mammography (CEM): Right Breast. Recombined CEM cranio-caudal (CC) and mediolateral oblique (MLO) views demonstrate a 4.4-cm moderately enhancing mass in the right mid-outer quadrant with associated linear non-mass enhancement, corresponding to the biopsy-proven invasive carcinoma of no special type (NST).

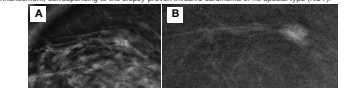


Figure 2A-B: Magnification view. The same right mid-outer quadrant mass at that is associated with amorphous calcification in linear distribution and corresponding CEM recombined image demonstrating high conspicuous mass and linear NME

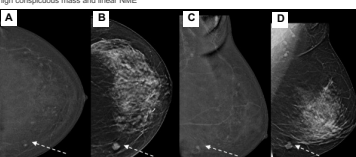
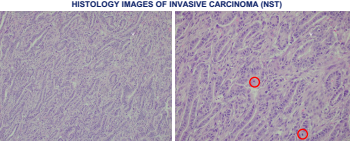


Figure 3A-D: Contrast-Enhanced Mammography (CEM): Left Breast. Recombined CEM CC and MLO views demonstrate a 1.4-cm irregular enhancing mass in the lower inner quadrant corresponding to the biopsy-proven mucinous carcinoma.



Figure 4A-B: Ultrasound images of left breast 7 o'clock lesion. 1.4-cm lobulated hypoechoic lesion corresponds to left lower inner mass on mammogram. This is biopsy-proven mucinous carcinoma.



Invasive carcinoma, NST, 40x magnification: Malignant cells are infiltrating in irregular tubules and fascicular pattern, associated with desmoplastic stromal reaction.

Invasive carcinoma, NST, under 100x magnification: The malignant cells exhibit moderate to marked nuclear pleomorphism, vesicular nuclei, prominent nucleoli and pale cytoplasm. A few mitotic figures are seen (red circles).

CASE 2: IPSILATERAL Synchronous tumour

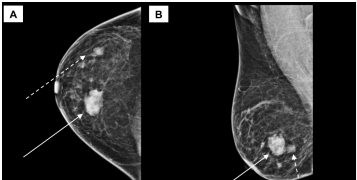


Figure 5A-B: Digital Mammography Right cranio-caudal (CC) and mediolateral oblique (MLO) mammograms demonstrate two irregular high-density masses in the right breast, corresponding to synchronous invasive carcinoma NST (3 o'clock, solid arrow) and mucinous carcinoma (8 o'clock, dashed arrow).

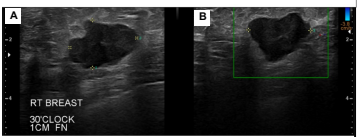


Figure 6A-B: Ultrasound images of right 3 o'clock lesion. Targeted ultrasound demonstrates a 2.2-cm irregular hypoechoic mass with irregular margin. Ultrasound-guided core biopsy confirmed invasive carcinoma of no special type (NST).

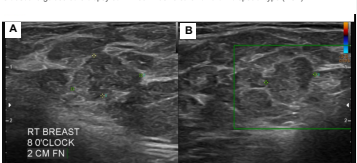
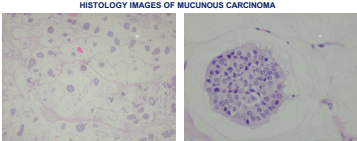


Figure 7A-B: Ultrasound images of right 8 o'clock lesion. Targeted ultrasound demonstrates a 1.2-cm isoechoic lesion with microlobulated margins. Ultrasound-guided core biopsy confirmed mucinous carcinoma.



Mucinous carcinoma, 200x magnification: Malignant tumour cells are seen floating within mucin pool.

Mucinous carcinoma, 400x magnification: A cluster of malignant cells displaying low grade nuclear features, with absence of myoepithelial layer, floating within mucin pool.

CONCLUSION / TAKE-HOME MESSAGE

SUMMARY OF MULTIMODALITY IMAGING AND HISTOPATHOLOGICAL FEATURES OF SYNCHRONOUS BREAST CARCINOMAS

Feature	Case 1 - Right Breast (NST)	Case 1 - Left Breast (Mucinous Carcinoma)	Case 2 - Right 3 o'clock (NST)	Case 2 - Right 8 o'clock (Mucinous Carcinoma)
Synchronous tumour	Bilateral synchronous tumour		Ipsilateral synchronous tumour	
Presentation	Screen-detected	Screen-detected	Diagnostic (palpable lump)	Diagnostic (palpable lump)
Clinical finding	Non-palpable	Non-palpable	Palpable	Palpable
Mammography	Equal-density mass associated with linearly distributed amorphous calcifications (MOQ)	Lobulated equal-density mass (LQ); separate UOQ architectural distortion with calcifications (benign)	Irregular spiculated high-density mass	Smaller irregular high-density mass in the lower outer quadrant
Ultrasound	No sonographic correlate (occult)	1.2-cm lobulated hypoechoic lesion at 7 o'clock	1.6 × 2.3 × 2.1 cm irregular hypoechoic lesion	0.8 × 1.3 × 0.9 cm isoechoic lesion with few microlobulations
CEM	Moderate homogeneous enhancing mass (4.4 cm) with associated linear non-mass enhancement	Irregular enhancing mass (1.4 cm) with a focal non-enhancing component; separate low-level heterogeneous enhancement (3.1 cm) corresponding to benign architectural distortion	Not performed	Not performed
Biopsy guidance	Stereotactic-guided	Ultrasound-guided	Ultrasound-guided	Ultrasound-guided
Histopathology	Invasive carcinoma NST	Mucinous carcinoma	Invasive carcinoma NST	Mucinous carcinoma
ER	90%	100%	20%	20%
PR	80%	80%	10%	10%
HER2	Equivocal	Negative	Negative	Negative
Final pathological stage	pT1bN0	pT1cN0	pT2N0	pT1bN0

KEY LEARNING POINTS

- ☐ **Synchronous tumours can be bilateral or ipsilateral.** Different histological subtypes may coexist within the same presentation.
- ☐ **Mucinous carcinoma has variable imaging appearances.** Benign-appearing morphology does not exclude malignancy.
- ☐ **Multimodality imaging is essential.** Independent assessment and biopsy of suspicious lesions support accurate staging and planning.

CONCLUSION

These cases underscore the importance of meticulous bilateral breast evaluation and radiologic-pathologic correlation. Recognition of the variable imaging manifestations of mucinous carcinoma, together with awareness that synchronous tumors of different histologic subtypes may coexist, is essential to avoid underestimation of disease extent, ensure complete lesion characterization, and facilitate appropriate surgical planning and multidisciplinary management.

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Author Contributions
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Dr. Asnani Mohd Ramli: Radiological review, supervision, and guidance.
Dr. Nur Dini Jalaludin: Histopathological review and contribution of pathology images.